

Initial Report of Group BLOCK UG13 “instruction set”

Block Model Compression Algorithm

Team members:

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Project Vision:

- Our project aims to develop an efficient compression algorithm tailored for block models, which is especially effective for low colour resolution images in 3D and is often used in geological mapping. Compressing these block models can be challenging, with vast arrays of color data that demand effective compression techniques. Scrum methodologies and agile frameworks will be employed to satisfy all requirements from the client, Maptek, a mining industry service provider.
- Our project vision is to improve speed and compression performance, but also balance high compression ratios with rapid processing to achieve an effective and useful block model compression algorithm.
- In short term, our goal is to reduce data size without any loss of data by grouping neighboring pixels into larger parent blocks and encoding remaining regions as collections of rectangular areas. In long term, we plan to significantly enhance compression performance. We will explore both CPU and GPU techniques to optimize processing speed and compression efficiency.
- The project's restrictions include output format, input/output size, handling uncompressed input and producing compressed output while ensuring no loss of data. Whereas the constraints include the users' hardware specification, such as the memory and the number of threads available, and implementation of GPU.

Customer Q&A:

Q: Do we need to submit with zero errors?

A: You can have errors as long as it is understandable.

Q: Can we compress the image so that the z-axis is 1?

A: The block object you compress must be 3D.

During the kickoff meeting, Simon Ratcliffe did a presentation on the project, which cleared up plenty of our questions related to the project. We learnt about Maptek, a multinational technology company based in Adelaide that develops world-leading software to help miners make better decisions. Also, we acknowledged that the algorithm developed by this project is very meaningful, has more practical needs, and can be extended to many fields. Additionally, it gave us a clear and concise understanding of the project requirements to begin with: to implement a stream processor, a new compression algorithm, and to submit an exe program. The algorithm has the requirement of real-time and compression ratio. We can iterate to improve our algorithm. We reflected on the kickoff meeting and realized that better organization of the questions and answers can be made. This can avoid confusion when the questions are being addressed and allow faster transfer of knowledge between teammates. Moreover, we learned that it was crucial to conduct thorough research before attending the kickoff meeting to ensure that we make the most use out of the meeting by addressing all the questions we can think of.

Users:

We identified two types of users for our solution - the developers who will implement the program and the clients who will use the program to compress their block models. The developers' role is to implement the program efficiently and effectively, while the clients' role is to use the program to compress their block models accurately and quickly in a specified format.

Software Architecture:

- We plan to use a modular architecture for our application, where each module performs a specific task. We will use object-oriented programming principles to design our modules, and we will use a layered architecture approach to separate concerns. We chose this architecture because it allows us to easily modify and extend our application in the future.

Part of the system is modeled as follows:

- **Context model**

Input data: The input block model data is provided through standard input (such as a text stream) and includes information such as the size of the block model, parent block size, and label table.

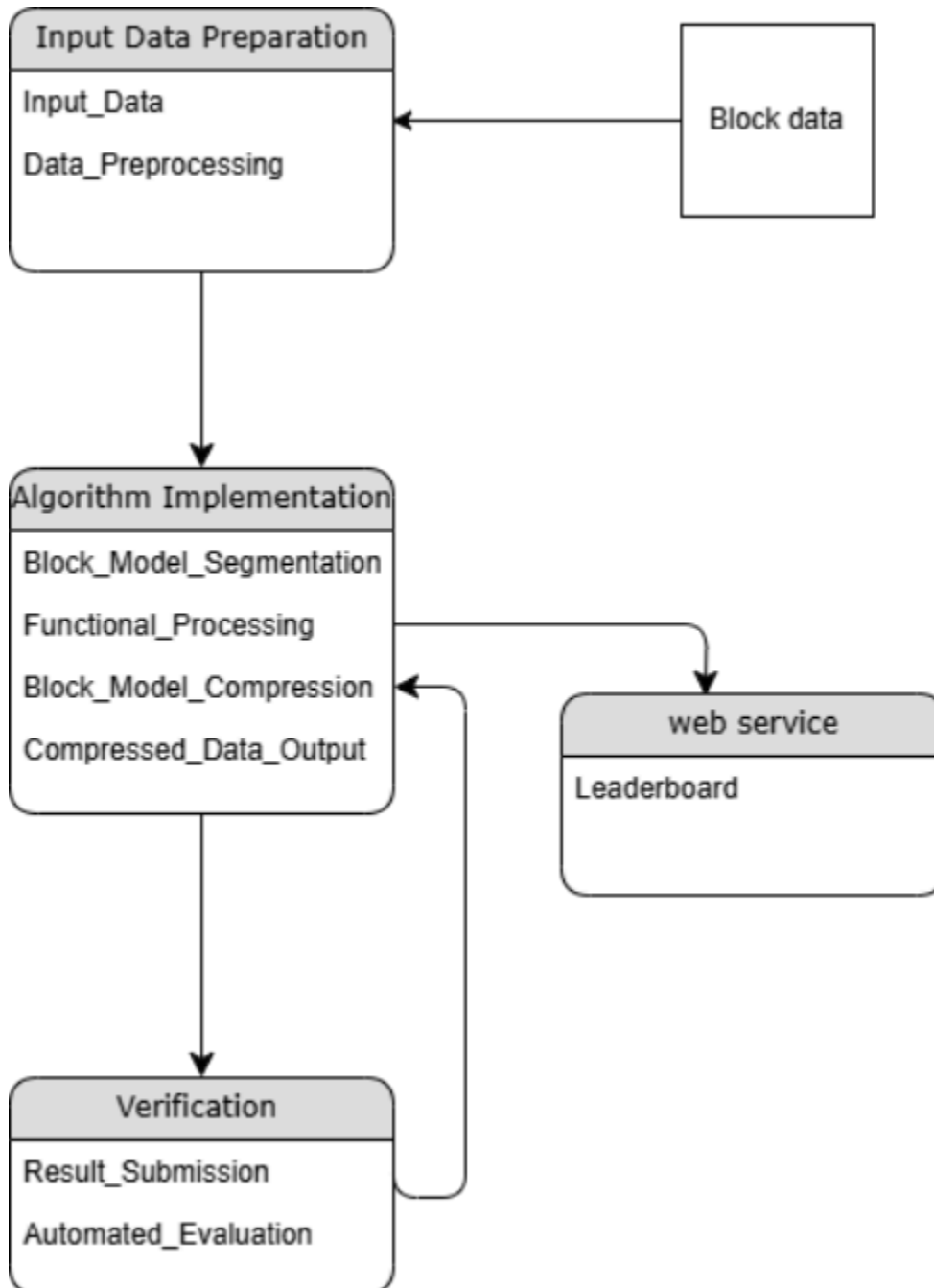
Algorithm implementation: It is necessary to implement a program that can read uncompressed input data and generate compressed output data without any information loss.

Validation Service: A small map of Australia that can be used to test developed algorithms.

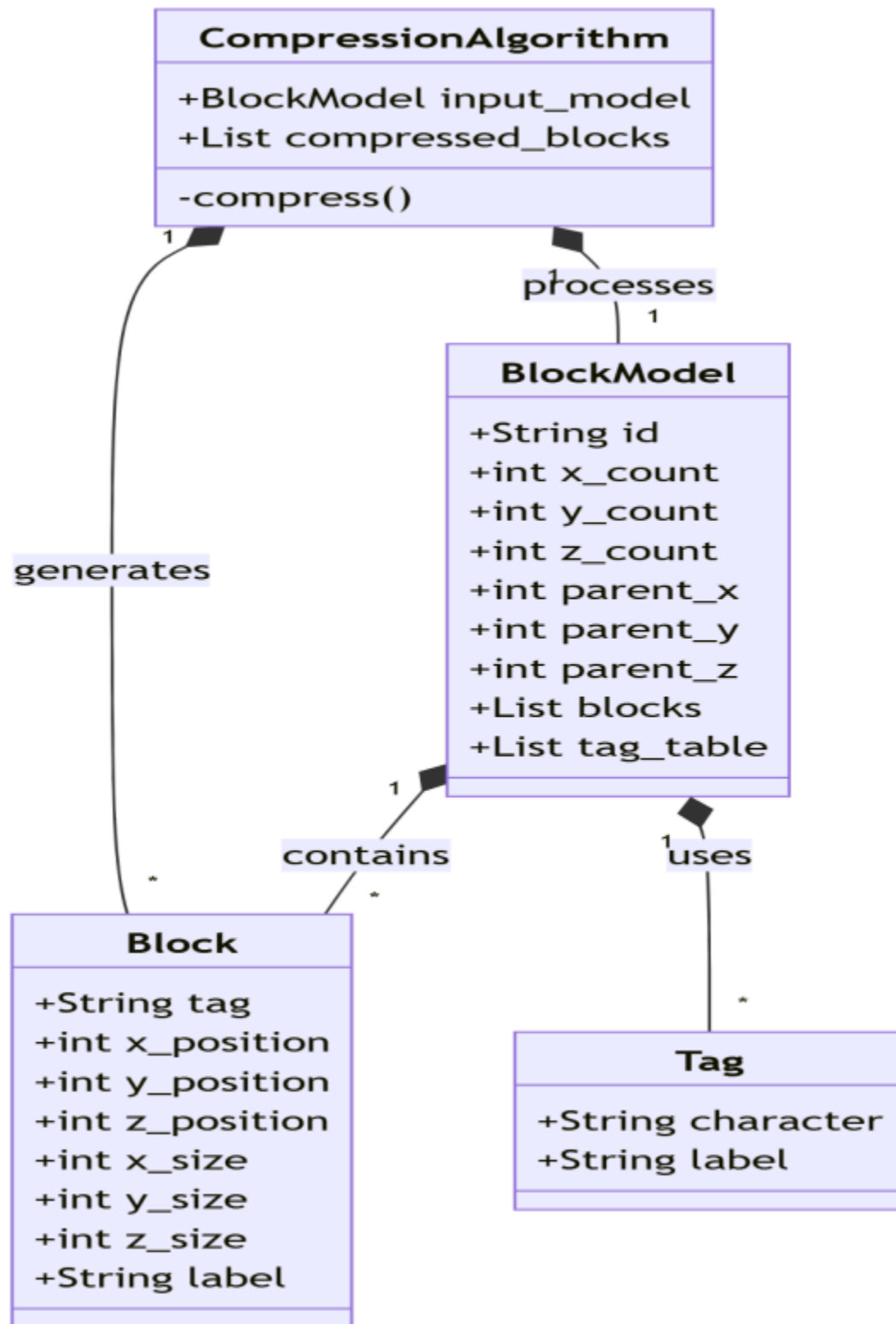
They need to upload their program to a website verification service.

The validation service performs these procedures and scores them based on processing speed and compression performance.

- Data-flow Models



- Object Models



Tech Stack and Standards:

Programming language:

- **C++:**(for high-performance system-level programming, particularly suited for developing compression algorithms that handle large data and complex computations.)
- **Python:**(will be utilized in conjunction with CUDA for GPU programming to accelerate algorithm execution)

Frameworks:

- Agree on tools for communication and development (e.g., Slack, IDEs, etc.)

Communication tools: Discord, Wechat, Microsoft Team

Development tools: VS Code, Github

- Agree on coding standards

We adhere to the Google C++ Style Guide and follow the Uncle Bob clean code guideline(<http://cleancoder.com/products>). focusing on meaningful naming, clear documentation, and performance optimization.

- Provide justifications for your choices

Focusing on meaningful naming, clear documentation, and performance optimization, and makes it easier for other developers to understand and modify.

Group Meetings and Team Member Roles:

- How frequently and for how long do you intend to (virtually) meet within your group?

Weekly integration of problems within the group (Every Sat). Follow existing problems in discord / MT / Wechat.

- What preparations should the group plan for before retrospective meetings?

Review: Review the task and feedback from previous retrospectives to assess progress and follow-up on any unresolved issues. The Scrum Master will also review what he did (manage) during that sprint.

Gather Data and Feedback: Collect relevant data and feedback from the sprint or project. This could include the performance of the algorithm, team feedback, and any issues.

Prepare Tools: Set up any tools or materials needed for the meeting, such as a notebook, PowerPoint. Make sure everything is in the github project file, so that the tutor can see our progress.

Schedule: Ensure the meeting is scheduled at a convenient time for all team members and send out reminders.

Define Objectives: Clearly define the goals of the retrospective, make a clear agenda for the meeting, outlining the topics to be discussed. Understanding what we want to achieve, such as reducing compression loss or speed improvement.

Planning: Plan our next sprint, such as set up a goal, schedule and divide our work. The next Scrum master needs to start planning how he will manage the team.

- How will the group communicate with the tutor?

The scrum master of our team then communicates with the client (tutor) on a weekly basis.

- Please name the Scrum Masters for each sprint (each team member can only be the Scrum Master for up to one sprint, except specific circumstances approved of by tutor).

Scrum Master(1st sprint): a1837374 a1837306

Scrum Master(2nd sprint): a1824243 a1837254

Scrum Master(3rd sprint): a1832623 a1860970

Scrum Master(4th sprint): a1818124

Scrum Master(5th sprint): a1837374 a1837306

- **Snapshot:**

- https://docs.google.com/document/d/1BQwtDJEDOebAzdsqo15jAZU6JseTNzHEuVdz_Kdqpyc/edit#heading=h.e9xtu5majca0